a) Inline Pump

b) Distributer Pump

6.6 FEED PUMP

Feed pump is another pump arrangement located between the diesel fuel tank and the filter in the diesel **Fuel Injection Pump** (FIP) block. The drive for the fuel feed pump is given from the camshaft of the fuel injection pump by cam or eccentric. The fuel feed pump is designed to operate manually by hand for purging the air in the fuel line when air bubbles are present. This fuel feed pump is classified into two types based on the delivery of fuel continuously or intermittent while the fuel injection pump is running. They are

1. Single acting pump
2. Double acting pump

6.6.1 Single Acting Pump

Construction

This type of pump is mounted on the body of the fuel injection pump. This single acting pump is driven by the cam or eccentric of the fuel injection pump's cam shaft. The components such as roller tapper, pressure spindle and plunger are in contact with each other and placed inside the pump body. For emergency purpose, hand priming device is used to increase the pressure of fuel. Figure 6.6.1 shows the Single Acting Pump.

Working Principle

When the engine is started the drive is received from the engine's crank shaft and given to the FIP cam shaft through timing gears. Now the cam shaft of the FIP rotates and while rotating the roller tappet the feed pump is actuated by the cam or eccentric in the cam shaft. Hence the plunger is actuated by the action of the roller tappet. Pressure spindle is now actuated by the plunger. Therefore, the fuel in the pressure chamber is pressurized and sent to the FIP. At the same time the high-pressure fuel developed in the FIP is sent to the nozzle (injector) by the FIP. The quantity of fuel delivered is equal to swept volume or equal to stroke length. The delivery passage is closed by the pressure spindle with the spring force. This is the way how the single acting pump works.

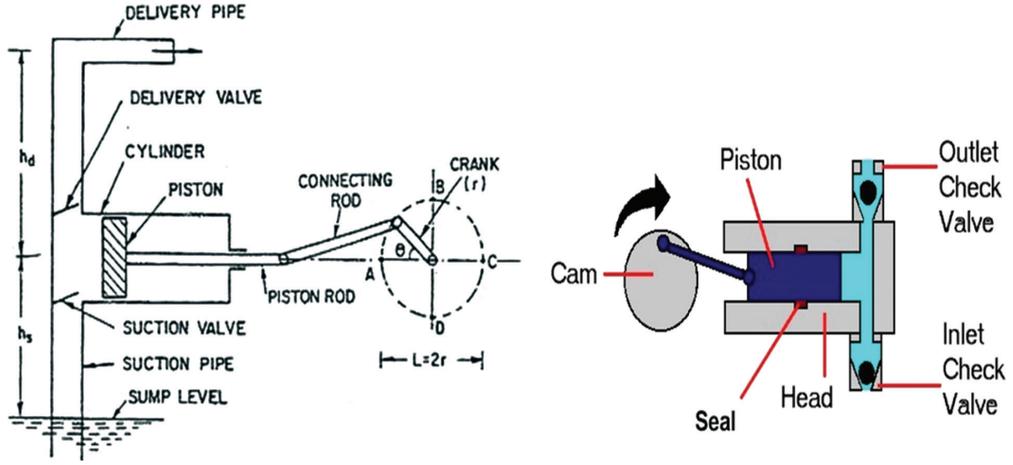


Figure 6.6.1 Single Acting Pump

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